MIL-C-60456 (MU)
27 October 1967

MILITARY SPECIFICATION

CARTRIDGE, 105MM, HE,
ROCKET ASSISTED, XM548
ASSESSORIES FOR,

1. SCOPE

1.1 This specification covers the accessories required in loading the rocket motor for the 105MM Rocket Assisted Cartridge, XM548.

2. APPLICABLE DOUCMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-14F - Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-A-2550 - Ammunition and Special Weapons, General Specification for.
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of.

STANDARDS

FEDERAL

Fed. Test Method STD. No. 406 - Plastics, Method of Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Quality Assurance Terms and Definitions.

FSC: 1315

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MIL-STD-453 - Inspection, Radiographic.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1169 - Packaging, Packing, and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ARMY

9212393 - Insulation Warhead.
9212396 - Insert, Nozzle.
9212397 - Insulation, Motor Body.
9212405 - Ring Seal.
9212406 - Sleeve, Grain Support.
9212407 - Closure, Igniter.
9212408 - Cushion Grain.
9212409 - Cap, Selector, Rocket.
9212410 - Gasket.
9212411 - Filler, Nozzle.
9212413 - Washer.
9224966 - Box, Fiber For Ring Seal For Cartridge, 105MM, HE, RAP XM548 (Metal Parts).

PUBLICATIONS

ARMY

ET-9212393 - Equipment Tabulation.
ET-9212396 - Equipment Tabulation.
ET-9212397 - Equipment Tabulation.
ET-9212405 - Equipment Tabulation.
ET-9212406 - Equipment Tabulation.
ET-9212408 - Equipment Tabulation.
ET-9212409 - Equipment Tabulation.
ET-9212410 - Equipment Tabulation.
ET-9212413 - Equipment Tabulation.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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OTHER PUBLICATIONS

AMMERICAN SOCIETY FOR TESTING AND MATERIALS

D638-58T - Tensile Properties of Plastics.
D1238-57T - Measuring Flow Rate of Thermo Plastics by Extrusion Plastometer.
D1505-57T - Measurement of Density of Plastics by the Density of Gradient Technique.

(Application for copies should be addressed to the Ammercian Society for Testing Materials, 1916 Race Street, Philadelphia, Pa.)

3. REQUIREMENTS

3.1 Material - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts - The parts shall comply with all requirements specified on the applicable drawing and with all requirements specified in applicable specifications.

3.3 Molding defects - The molded components shall be free of porosity, foreign inclusions, ripples, cracks or surface imperfections.

3.4 Workmanship - All parts shall be fabricated and finished in a thorough workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, tears, holes, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

3.5 First Article Inspection - This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.6 Sleeve Grain Support

3.6.1 Physical Properties - The high density polyethylene sleeve shall conform to the physical properties listed in Table I. (See dwg. 9212474 for identification of manufacturers product).

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3.6.2 Reworked material.-No reworked material or reground scrap shall be utilized in molding sleeves under this specification. However, initial pelletizing of the molding composition used shall not be construed as reworking.

3.7 Nozzle Insert (See Figure 1)

Zone A - There shall be no voids in Zone A greater than 0.030 inch in diameter when tested as specified in 4.4.2.1.

Zone B - There shall be no voids greater than 0.060 inch in diameter. There shall be no voids greater than 0.030 inch in diameter which are closer than 0.060 inch apart when tested as specified in 4.4.2.1.

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TABLE I

COMPOSITION	A	B	C	D	E	F	G	H	J
Tensile Strength at yield (minimum P.S.I.)(Min)									
23 degrees centigrade(C)	2760	2060	2270	2770	2800	2700	2550	2800	2500
71 degrees (C.)	1100	1000	1000	1150	1050	1050	1050	1000	950
-54.3 degrees (C.)	6100	5300	5860	6400	6400	6300	6150	6500	6000
Elongation (Min) Percent									
23 degrees C.	10	11	13	12	9	10	11	10	10
71 degrees C.	13	38	39	38	24	11	33	15	12
-54.3 degrees C.	5	5	5	4	4	5	5	4	5
Modulus of Elasticity (X10 to the fifth) (PSI) (Min)									
23 degrees C.	1.40	1.00	1.04	1.44	1.60	1.40	1.38	1.44	1.30
71 degrees C.	0.25	0.19	0.22	0.30	0.28	0.25	0.27	0.24	0.90
-54.3 degrees C.	4.00	3.30	3.46	3.64	2.14	4.00	3.82	4.64	4.00
Melt Index (grams/10 minutes with 2160 g. Load Range, plus or minus 0.5 g)	2.9	1.2	1.2	1.0	1.0	1.0	0.76	0.7	1.56
Flow Ratio	7-10	12-23	9-13	12-17	10-15	10-22	12-20	12-17	12-19
Density g/ML(plus or minus .005	0.969	0.958	0.959	0.969	0.978	0.968	0.966	0.967	0.968
Thermal Stress Cracking Resistance	The sleeves shall not crack, split, warp, or break when subjected for 100 hours at 71 degrees C. when tested as specified.								

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4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.
h. Certificate that nozzle insert grain flow is as specified on the drawing.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.1.2 Government verification.-Using the contractor's written quality assurance procedure, this detailed specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals, all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 First Article Inspection

4.2.1 Submission.-The contractor shall submit a first article sample consisting of ten (10) nozzle inserts, ten (10) Cap, Rocket selectors, ten (10) Ring seals, ten (10) insulation, motor bodies from each cavity of each mold, ten (10) sleeve grain supports from each cavity of each mold, ten (10) closure igniters, from each set of dies, ten (10) cushion grains from each set of dies, ten (10) gaskets from each set of dies, ten (10) nozzles fillers from each cavity of each mold and ten (10) insulation warheads from each cavity of each mold or set of dies, as applicable. The sample shall be produced by the contractor or furnished by a supplier and shall have been manufactured using the same production processes, procedures, and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample tothe degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings/certificates of conformance for materials (see 4.1.1.1.e). All molded or punched components shall be identified as to cavity or die from which the part was produced. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. A first article sample or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

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4.2.2 Inspections to be performed - Components will be subjected by the Government to any or all of the examinations or tests specified in 4.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection - If any component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection Provisions

4.3.1 Lot formation. - The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for isnpection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. Changes to the process, specifications, or drawings, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery, during the contract period. Inspection lots of components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutaully convenient to both the contractor and the Government inspector. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

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4.3.2 Examination - Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using a AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Once agreement has been reached as to which method will be used (group or individual) that method shall be used for all of the classification of defects listed in this specification. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Ring, Seal (see dwg. 9212405).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.50 percent		
201.	Inside diameter	Gage	01001
202.	Length, min	Gage	01002
203.	Thickness	Gage	01003
204.	Evidence of poor workmanship (see 3.4)	Visual	01004

4.3.2.2 Gasket (see dwg. 9212410).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Thickness	Gage	02001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	02002

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4.3.2.3 Insert, Nozzle (see dwg. 9212396).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Throat diameter	Gage	03001
102.	Diameter at end of small diverging cavity	Gage	03002
103.	Concentricity of cavity at .06 location with outside diameter	Gage	03003
104.	Outside diameter	Gage	03004
105.	Total length	Gage	03005
106.	Surface finish improper	Visual	03006
107.	Chamfer missing	Visual	03007
Minor:	AQL 4.00 percent		
201.	Concentricity of throat with outside diameter	Gage	03008
202.	Depth to 0.396 basic diameter	Gage	03009
203.	Diameter at end of large diverging cavity	Gage	03010
204.	Concentricity of cavity at .12 location	Gage	03011
205.	Radius of small diverging cavity	Gage	03012
206.	Radius of large diverging cavity	Gage	03013
207.	Flat, max	Gage	03014
208.	Marking missing, misleading or illegible	Visual	03015
209.	Evidence of poor workmanship (see 3.4)	Visual	03016

4.3.2.4 Cap, Selector, Rocket (see dwg. 9212409).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Major:	AQL 0.65 percent		
101.	Inside diameter	Gage	04001
102.	Pitch diameter of thread	Gage	04002
103.	Concentricity of thread with inside diameter	Gage	04003
104.	Surface finish of inside diameter	Visual	04004
Minor:	AQL 4.00 percent		
201.	Minor diameter of thread, max.	Gage	04005
202.	Width of thread undercut	Gage	04006
203.	Diameter of thread undercut	Gage	04007
204.	Total length	Gage	04008
205.	Distance end to bottom of cavity	Gage	04009
206.	Distance bottom of cavity to intersection of thread chamfer and inside diameter	Gage	04010
207.	Knurl missing or incomplete	Visual	04011
208.	Chamfer on inside diameter missing	Visual	04012
209.	Chamfer on thread missing	Visual	04013
210.	Protective finish missing or incomplete	Visual	04014
211.	Marking missing, incomplete, misleading or incorrectly applied	Visual	04015
212.	Undercut fillet radius missing	Visual	04016
213.	Evidence of poor workmanship (see 3.4)	Visual	04017

4.3.2.5 Insulation, Motor Body (see dwg. 9212397).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Depth of depression at large diverging cavity minimum	Gage	05001
102.	Length of projection, minimum	Gage	05002
103.	Thickness	Gage	05003
104.	Parallelism of surfaces	Gage	05004
105.	Abrasive finish missing or excessive	Visual	05005
106.	Molding defects (see 3.3)	Visual	05006
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	05007

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4.3.2.6 Closure, Igniter (see dwg. 9212407).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4).....	Visual	06001

4.3.2.7 Cushion, Grain (see dwg. 9212408).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Thickness.....	Gage	07001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4).....	Visual	07002

4.3.2.8 Filler, Nozzle (see dwg. 9212411).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.00 percent		
201.	Center hole incomplete.....	Visual	08001
202.	Evidence of poor workmanship (see 3.4).....	Visual	08002

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4.3.2.9 Washer (see dwg. 9212413).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Flatness	Gage	09001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	09002

4.3.2.10 Insulation Warhead (see dwg. 9212393).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Ground surface finish missing or inadequate	Visual	10001
Minor:	AQL 1.00 percent		
201.	Flatness	Gage	10002
202.	Evidence of poor workmanship (see 3.4)	Visual	10003

4.3.2.11 Sleeve, Grain Support (see dwg. 9212406).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 100 percent		
101.	Total length	Gage	11001
Minor:	AQL 1.50 percent		
201.	Excess flashing	Visual	11002
202.	Molding defects (see 3.3)	Visual	11003
203.	Evidence of poor workmanship (see 3.4)	Visual	11004

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4.3.3 Testing

4.3.3.1 First Article Sample (see 4.2.1) - The samples submitted as the first article quantity shall be tested as specified below after having been examined for all drawing dimensions and requirements. Rejection and resubmission shall be in accordance with 4.2.3.

4.3.3.1.1 Sleeve Grain Support

4.3.3.1.1.1 Physical Properties - Code No. 11005 - Five test specimens for each of the temperatures specified in Table I shall be molded under the same conditions as will be used in regular production. The test shall be conducted as specified in 4.4.1.1. If any specimen fails to meet the tensile strength, elongation, or modulus of elasticity requirement of Table I, the first article sample shall be rejected.

4.3.3.1.1.2 Melt Index

4.3.3.1.1.2.1 Melt Index (See Table I) Procedure A - Major defect - Code No. 11006. - Five (5) Sleeves from each cavity shall be selected and tested in accordance with 4.4.1.2.1. If the average of the five (5) specimens fails to comply with the requirements of Table I, the first article sample shall be rejected.

4.3.3.1.1.2.2 Melt Index - Procedure B - Code No. 11007. - One test specimen from each of the five sleeves from each cavity used in 4.3.3.1.1.2.1 shall be selected and tested in accordance with 4.4.1.2.2 - The five determinations per cavity using the 10Kg load shall be averaged and the average used in calculating the flow ratio of 4.3.3.1.1.3.

4.3.3.1.1.3 Flow ratio - (see Table I) - Major defect - Code No. 11008 - The flow ratio shall be determined as specified in 4.4.1.3. If the flow ratio fails to meet the requirements of Table I the first article sample shall be rejected.

4.3.3.1.1.4 Density (See Table I) - Major defect - Code No. 11009 - One specimen from each of five sleeves from each cavity shall be used for this test. The test shall be conducted in accordance with 4.4.1.4. If any specimen fials to meet the requirements of Table I the sample shall be rejected.

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4.3.3.1.1.5 Thermal stress cracking resistance (See Table I) - Major defect - Code No. 11010. - Five sleeves from each cavity shall be selected for this test. The test shall be conducted as specified in 4.4.1.5. If any sleeve fails to meet the requirements of Table I the first article sample shall be rejected.

4.3.3.1.2 Nozzle Insert

4.3.3.1.2.1 Radiographic inspection - Major defect - Code No. 03017. - Each nozzle insert submitted for inspection shall be radiographed for voids in accordance with 4.4.2. If any piece fails to meet the applicable requirements the sample shall be rejected.

4.3.3.2 Regular Production

4.3.3.2.1 Nozzle Insert

4.3.3.2.1.1 Grain flow - A certificate of compliance shall be submitted with each lot that the grain flow of the nozzle insert is in accordance with the drawing. The contractor shall show evidence that adequate controls are being maintained in processing the nozzle insert from the billet to finished component to assure proper grain flow direction.

4.3.3.2.1.2 Radiographic Inspection - Major defect - Code No. 03017. - This test shall be conducted 100 percent. The test shall be conducted in accordance with 4.4.2. If any nozzle insert fails to meet the requirements of 3.7 that piece shall be rejected.

4.3.3.2.2 Molded and stamped components.-After production of the quantities specified in Table II, 10 pieces per cavity per mold or set of dies shall be consecutively produced and inspected for all drawing and specification requirementsand verified by a Government Inspector. If any part fails to meet all drawing or specification requirements, production shall be stopped and the mold or die corrected. During production, the molding procedures shall be monitored for uniformity of molding practices. Suspected deviations in mold product shall be examined to determine if corrective action or mold repair is required.

TABLE II

Component	Quantity Per Cavity or Die (as Applicable)
Insulation, warhead	20,000
Insulation, Motor Body	10,000
Nozzle, Filler	20,000
Closure, Igniter	20,000
Cushion, Grain	20,000
Washer	20,000
Sleeve, Grain Support	20,000

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4.3.3.2.3 Sleeve, Grain Support

4.3.3.2.3.1 Melt Index (See Table I) - Procedure A - Major defect - Code No. 11006. - Three sleeves shall be selected from each lot and tested in accordance with 4.4.1.2.1. If the average of the three specimens fails to comply with the requirements of Table I, the lot shall be rejected.

4.3.3.2.3.2 Melt Index - Procedure B - Code No. 11007 - One test specimen from each of the sleeves used in 4.3.3.2.3.1 shall be tested in accordance with 4.4.1.2.2. The three determinations using the 10Kg. load shall be averaged and the average used in calculating the flow ratio of 4.3.3.2.3.3.

4.3.3.2.3.3 Flow Ratio (See Table I) - Major defect - Code No. 11008. - The flow ratio shall be determined as specified in 4.4.1.3. If the flow ratio fails to meet the requirements of Table I, the lot shall be rejected.

4.3.3.2.3.4 Density (See Table I) - Major defect - Code No. 11009. - Three sleeves shall be randomly selected from each lot for this test. The test shall be conducted in accordance with 4.4.1.4. If any specimen fails to meet the requirements of Table I, the lot shall be rejected.

4.3.3.2.3.5 Thermal stress cracking resistance (See Table I) - Major defect - Code No. 11010. - Five sleeves shall be selected at random from each lot for this test. The test shall be conducted as specified in 4.4.1.5. If any sleeve fails to meet the requirements of Table I the lot shall be rejected.

4.3.4 Inspection Equipment - Equipment Tabulation number ET-9201930 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation. - Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

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4.3.4.2 Supply and maintenance. - Supply and maintenance of the equipment listed on the ET shall be in accordance with specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection equipment. The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test Methods and Procedures

4.4.1 Sleeve, Grain Support

4.4.1.1 Tensile strength, elongation, and modulus of elasticity.

4.4.1.1.1 Five of the molded test specimens shall be conditioned for a minimum period of six hours at 23 degrees centrigrade, plus or minus 1 degree C. The test shall be conducted in accordance with ASTM-D-638-58T or Method 1011 of Fed. Test Method Std. 406. The speed of testing shall be speed B of AMSTM-D-638-58T.

4.4.1.1.2 Five of the molded test specimens shall be conditioned for a minimum period of one hour at a temperature of 71 degrees centrigrade, plus or minus 1 degree C. and tested as specified in 4.4.1.1.1.

4.4.1.1.3 Five of the molded test specimens shall be conditioned for a minimum period of one hour at a temperature of minus 54.3 degrees Centrigrade, plus or minus 1 degree C. and tested as specified in 4.4.1.1.1.

4.4.1.2 Melt Index

4.4.1.2.1 Procedure A - One specimen shall be selected from each of the sample sleeves. The test shall be conducted in accordance with ASTM-D-1238-57T, Condition E.

4.4.1.2.2 Procedure B - One specimen shall be selected from each of the same sleeves used in 4.4.1.2.1. The test shall be conducted as specified in 4.4.1.2.1 except that a 10Kg load shall be substituted for the 2160 gram load specified in ASTM-D-1238-57T.

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4.4.1.3 Flow Ratio - The flow ratio shall be determined by comparing the values obtained in 4.4.1.2.1 and 4.4.1.2.2. (The flow ratio is defined as the ratio of the flow rate in a standard melt index with a 10Kg piston loading, 4.4.1.2.2, to the standard melt index, 4.4.1.2.1).

4.4.1.4 Density - The density of the specimens shall be determined in accordance with ASTM-D-1505-57T. One test specimen shall be selected from each of the same sleeves used to obtain specimens for 4.4.1.3. Specimens shall not be annealed or otherwise treated so as to change the crystallinity. (See 6.4)

4.4.1.5 Thermal stress cracking resistance - The sleeves shall be conditioned for 100 hours at 71 degrees Centrigrade, plus or minus 1 degree C. Immediately following the 100 hour time period each sleeve shall be removed individually and shall be visually examined while warm (see Table I). After cooling to 23 degrees Centrigrade plus or minus 1 degree C., the sleeves shall each be cut open longitudinally and the interior surfaces examined visually for evidence of cracks, splits, warping or other deformations.

4.4.2 Nozzle Insert

4.4.2.1 Radiographic test.-The nozzle insert shall be radiographed with the large diverging cavity down and the radiographic beam parallel to the longitudinal axis. Radiographic inspection shall be conducted in accordance with MIL-STD-453. The penetrameter shall be designed so as to assure a 2 percent sensitivity. The film density shall be 1.50 minimum.

5. PREPARATION FOR DELIVERY.

5.1 Preservation and packaging

5.1.1 Level C - Preservation and packaging of accessories other than Ring, Seal shall be in accordance with MIL-STD-1169.

5.1.2 Level C - Preservation and packaging of Ring, Seals shall be in accordance with dwg. 9224966.

5.2 Packing

5.2.1 Level C - Packing of accessories other than Ring, Seal shall be in accordance with MIL-STD-1169.

5.2.2 Level C - Packing of Ring, Seals shall be in accordance with dwg. 9224966.

5.3 Marking - Marking shall be in accordance with MIL-STD-1169.

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6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. - The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists. - Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes. - The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs. - Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use

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these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs. - Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs. - All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND4. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

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6.4 Density specimens. - Saw speeds in excess of 500 to 600 lineal feet per minute have been found to be unsatisfactory as excess speed can essentailly melt the material and greatly change density.

6.5 Submission of First Article Samples. - Instructions as to location for the evaluation of First Article samples shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to Commanding Officer, Picatinny Arsenal, Dover, New Jersey, ATTN: SMUPA-NI.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project No. 1315-A-571

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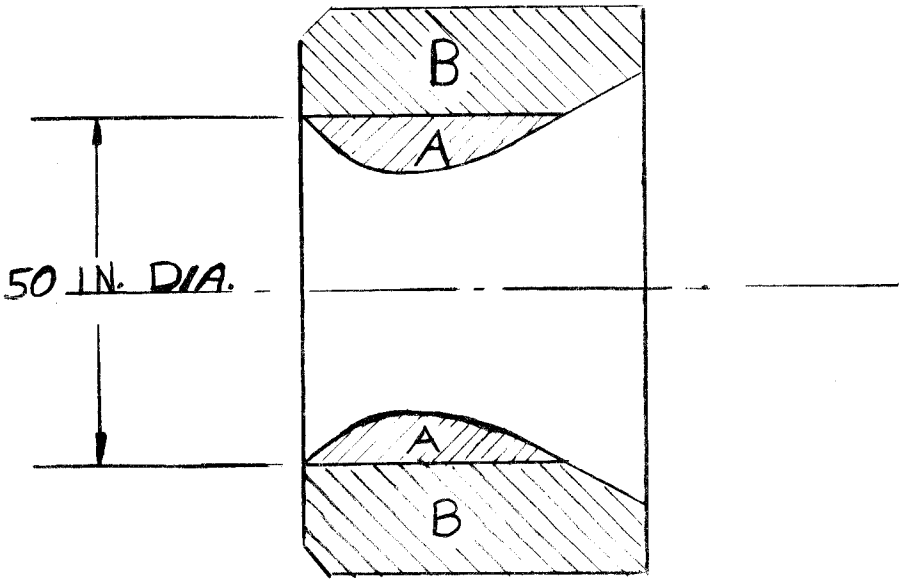


FIGURE 1

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